

DATA-DRIVEN SIMULATION-BASED ANALYSIS OF COLLABORATIVE BUSINESS PROCESSES IN DISTRIBUTED ENVIRONMENTS

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ABSTRACT

This paper proposes a novel approach to the predictive analysis of business processes in distributed and collaborative environments. The work introduces the joint use of Process Mining approaches and Distributed Simulation techniques to support the analysis of business process collaborations, which are complex and distributed flows of activities enacted by cooperating participants to achieve a common objective. Specifically, the proposed method enables each participant to build a simulation model which describes the process collaboration from its specific point of view. The method also uses a low-code development paradigm to build an independent simulator, starting from the simulation model. The various simulators are then deployed and executed in a federated and distributed simulation to analyze the collaborative scenario addressed. The method is finally applied to a concrete case to highlight the potential benefits of the proposed approach.

Keywords: Predictive Process Mining, Distributed Simulation, Model Transformations, Business Processes, BPM)

1 INTRODUCTION

The analysis of Business Process (BP) collaborations poses significant challenges in the context of business process management. These collaborations involve flows of activities that extend beyond the boundaries of a single organization, capturing interactions among multiple entities that work together to achieve a common goal [1].

The ability to properly analyze and evaluate the potential for collaboration is essential for organizations planning to share their resources in collaborative workflows. In this respect, Modeling & Simulation (M&S) is recognized as a valuable option to quantitatively analyze BPs in terms of performance and effectiveness [2]. Any M&S effort requires the definition of a sufficiently detailed model that captures all relevant aspects of the scenario of interest, along with accurate quantitative information, to build an executable simulation. Unfortunately, the peculiarities of the addressed collaborative scenario significantly impact the M&S analysis strategy to be adopted.

BP collaborations rely on the exchange of information among distributed participants, who aim to create value through cooperation while maintaining the confidentiality of their internal business processes' structure and implementation. Such a collaborative scenario focuses on the specification of appropriate interfaces and a communication protocol that rules the information exchange. Therefore, each participant has a partial knowledge of the whole process collaboration, limited to the activities he is responsible for, and lacks sufficient information to model the entire process flow. Moreover, conventional discrete-event simulation tools often fall short in addressing the inherently distributed, heterogeneous, and interoperable nature of process collaborations. From a modeling perspective, Process Mining (PM) tools can be effectively adopted to discover process models from event logs recorded by process-aware information systems (PAIS) that

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each participant exploits to execute its own BPs [3]. In [4], a *Predictive Process Mining (PPM)* approach has been proposed to support BP analysis by jointly exploiting Process Mining and M&S-based techniques. Specifically, PPM leverages PM methods to support the development of a simulation model by analyzing existing system logs. The simulation model is then implemented and executed to obtain predictions about the future behavior of the process under study, with simulation results complementing the insights gained from log analysis [5]. However, the limited information available to each partner for PM discovery makes this task harder to accomplish in the context of BP collaborations.

From the simulation implementation perspective, Distributed Simulation (DS) approaches, such as those based on the High-Level Architecture (HLA) standard [6], naturally meet the BP Collaboration requirements. HLA allows for the specification and implementation of a distributed simulation as a federation of simulation components (i.e., federates in the HLA terminology), each representing how individual participants contribute to a process collaboration. Federates act as black-box components that exchange agreed-upon information without revealing their internal implementation details.

Building on these premises, this paper introduces a framework for predictive analysis and process enhancement in distributed and collaborative environments. Specifically, it proposes a novel approach for *distributed simulation-based PPM* of BP collaborations. The method builds on traditional PPM analysis methods and takes a further step by leveraging the Distributed Systems (DS) extension for the eXtensible Event Stream (XES) format, as introduced in [7]. The DS extension, specifically tailored for distributed process simulations, effectively captures interactions among distributed participants while preserving the confidentiality of their internal processes.

The remainder of the paper is structured as follows. Section 2 clarifies the positioning of this contribution with respect to the previous work of the authors and outlines the state of the art. Section 3 illustrates the particular challenges that this work aims to address and describes in detail the proposed method for the DS-based PPM of BP collaborations. Section 4 discusses the application of the proposed method to a concrete case. Finally, Section 5 gives closing remarks.

2 BACKGROUND

Various contributions can be found in the literature that investigate approaches and technologies to analyze process collaborations. In this respect, this section is structured as follows. Section 2.1 briefly summarizes authors' previous work, which addresses: a) *low-code* and *model-driven* methods to support the development of distributed simulation systems, and b) predictive process mining. Moreover, Section 2.2 reviews the existing literature in the field of PM and DS, specifically with regard to the analysis of process collaborations.

2.1 Past Contributions and Paper Positioning

The analysis of collaborative processes has been the subject of our previous work, which addresses this topic from the following perspectives:

- In [8], a framework is defined based on low-code approaches for the analysis of BP collaboration through distributed simulation. Specifically, this work focuses on defining methods for the (semi-) automatic generation of the necessary artifacts for implementing a simulation based on HLA and the Java-based eBPMN framework. The proposed work has then been extended in [9] by discussing the application of the method to a concrete case study, implemented through a distributed and heterogeneous simulation that includes both eBPMN- and DEVS-based federates.
- In [4], the idea of jointly using Process Mining (PM) and simulation in a Predictive Process Mining (PPM) approach is introduced. This contribution, which focuses on the use of local simulation, also

highlights potential developments of PPM, such as increasing the level of automation and introducing DS technologies to leverage the distributed context of process collaborations.

- in [7], an extension of the XES standard is proposed, specifically designed for the discovery of BP collaboration models to support DS.

As the extension plays an essential role in this work, the additional attributes it provides are summarized hereafter. For a complete specification of the DS Extension metamodel, interested readers are referred to the cited literature. The **entity** attribute extends the information captured at the log level, that is, the information that characterizes the execution of the whole process. It is a `string` attribute that identifies the name of an entity involved in the process collaboration. It is further specified by two elements: *host*, which denotes a `string` attribute providing the IP address or the hostname, and the `boolean` attribute *isRemote*, which denotes if the entity is `local` or `remote`. Moreover, additional attributes further extend the information captured for each event recorded in the log:

- **localEntity**: optional attribute of `string` type, which provides the name of the local entity in a given communication. The attribute value must refer to one of the entities specified at the log level.
- **remoteEntity**: attribute of `string` type, which provides the name of the remote entity in a given communication. The attribute value must refer to one of the entities specified at the log level.
- **communicationType**: attribute of `string` type, which denotes direction of the communication flow. It can assume one of the following value: `send`, `receive`, `send-and-receive` or `receive-and-send`.
- **data**: attribute of `string` type, which identifies the communication payload, i.e., the data exchanged between two entities. As discussed in Section 3.1, an additional UML class diagram shall be provided to fully describe the exchanged information. This model is referred to as the *Federation Data Model*.

The framework proposed in this paper addresses the challenges identified in [4], particularly by improving the automation degree of simulation model specification and implementation, as described in Section 3.2. Moreover, the DS Extension is fully integrated into the proposed method to effectively support the PPM of BP collaborations.

2.2 Related Work

The originality of the proposed work lies in the identification of an approach to analyze BP collaborations through the combined use of PM and DS-based techniques. To the best of our knowledge, no similar works exist in the literature. However, the various topics addressed in this work have been widely investigated.

The discovery of process models in the context of BP collaborations is a particularly challenging topic [10, 11]. Specifically, in [10], a technique is proposed to discover collaborative inter-organizational processes. This work leverages an extension of the XES standard to address the challenge of unifying data scattered across various system logs. The paper also discusses the concrete implementation of the proposed approach as a plugin for the ProM framework. In [11], a different approach is proposed that focuses on the messages exchanged between parties. The authors introduce an analyzer capable of pairing each sent and received message between two participants, using this information to derive the resulting collaborative process model. Unlike such approaches, this contribution's objective is not limited to the identification of a method for discovering collaborative process models. It also aims to provide a comprehensive methodology that might effectively support the simulation of BP collaborations in a distributed scenario.

The joint adoption of PM and simulation-based techniques to support the BP management effort has been explored in various works, such as [5, 12]. In these approaches, log analysis is used to discover the information necessary to specify and parameterize a local simulation model. This model utilizes prior knowledge of a system or process to enable simulation-based studies of alternative scenarios.

Based on these significant contributions, this work aims to further advance the analysis of BP collaborations. The proposed method overcomes the limitations of traditional PM approaches, which focus on single-pool BPs. Moreover, by leveraging low-code programming principles and DS techniques, it effectively supports a comprehensive simulation analysis through the execution of a federation of simulators, each independently developed by the relevant participants.

3 PPM-BASED ANALYSIS OF PROCESS COLLABORATIONS

This section illustrates the proposed method to carry out a PPM-based analysis of BP collaborations that specifically leverages DS techniques. While Section 3.1 clarifies the context and rationale of the method, Section 3.2 illustrates in detail the various activities in which it is structured, along with relevant implementation details.

3.1 Process Collaborations: Characteristics and Challenges

A BP collaboration can be viewed as a *complex workflow* involving multiple organizations. Each participant operates autonomously, relying on its own execution infrastructure. There is no single repository that collects all events captured during the execution of the collaborative workflow: events' data are scattered across participants' PAIS. As a result, there are n independent event logs, each capturing the execution of the process from the perspective of a single participant, as illustrated in Figure 1. In other words, each system log records only the activities performed by the systems and resources of its corresponding participant.



Figure 1: Process collaboration scenario.

The objective of the proposed method is to enable each participant to develop an independent simulator, which is then deployed and executed in a federated and distributed simulation. Each federate can rely only on the (partial) information stored in its system log. In this respect, the additional information provided by the DS Extension allows participants to discover a process collaboration model that specifies:

- the flow of activities executed within the participant's domain;
- the data exchanged with other entities.

As shown in Figure 2, each participant analyzes its log to discover a BPMN collaboration model consisting of n pools, where only the pool associated with the given organization is fully detailed, while the remaining pools are represented as *black boxes*. This model is then used to develop, using an appropriate *low-code* approach, a federate that mimics the behavior of the participants. All federates are then executed in a federated and distributed simulation to obtain predictions about the entire collaborative process.

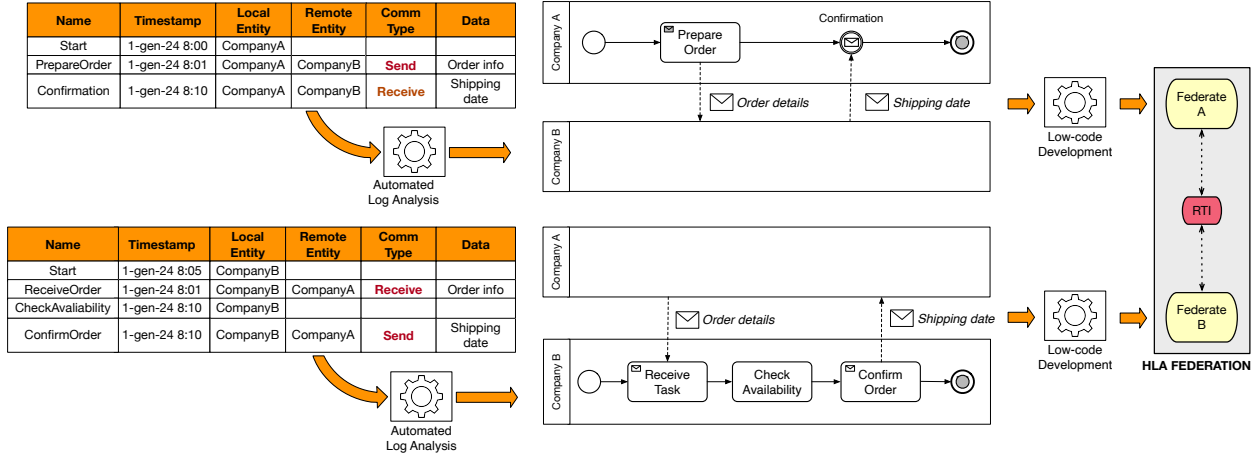


Figure 2: Rationale of the proposed method. Each participant analyzes its own log to build one independent simulator, which, in turn, is deployed in a federated and distributed simulation scenario.

3.2 Method Overview

This section illustrates the proposed method that enables the PPM-based analysis of process collaborations by jointly exploiting process mining techniques and distributed simulation technologies. In this regard, as underlined in Section 1, the proposed method adopts HLA, the reference standard in the distributed simulation field.

As described in Section 3.1, in the addressed scenario, multiple partners cooperate to achieve a common goal by implementing a collaboration process. In this respect, each participant executes a given flow of activities and exchanges data with other parties. The proposed method is based on the analysis of the event log of each participant in order to develop an autonomous federate that will be executed alongside the federates developed by other participants in a federated distributed simulation. Although the log analysis allows participants to discover the flow of executed activities and the communication protocol in terms of data entities exchanged among the involved parties, this information alone is not sufficient to support the federate development. In order to ensure interoperability of federates based on different implementation technologies, the implementation of an HLA federate requires the specification of:

- a data model, namely the Federation Object Model (FOM) [6], which describes the data entities that federates produce or consume during the federation execution. In HLA, such data entities must be typed by use of HLA datatypes, as prescribed by the Object Model Template (OMT) specification.
- the capabilities of each federate in producing or consuming the data entities specified by the FOM. Specifically, HLA adopts a *publish and subscribe* paradigm. A federate publishes the data entities it produces and subscribes to the data entities it requires for its computation.

From an implementation perspective, each federate must handle the data entities specified in the FOM appropriately. This requires implementing a mapping between HLA datatypes and the specific datatypes supported by the programming language in use. For instance, let us consider a federation simulating an Air Traffic Control System (ATCS). Among other data, let us assume that the ATCS must manage a list of incoming flights, where each element is a string representing a flight number. In the FOM, the *flights* list could be defined as an *HLAvariableArray* of elements typed as *HLAunicodeString*. In a corresponding Java implementation, these datatypes would be mapped to a collection of objects of type *String*.

In this respect, in the proposed approach, collaboration participants agree on a shared data model that describes the data entities exchanged by federates during the federation execution. This data model is specified as a UML model, referred to as the *Federation Data Model*. As explained below, the Federation Data Model will be used to automatically generate the FOM (Section 3.2.1) and to create the appropriate data structures in the federate implementation (Section 3.2.4).

The proposed method is illustrated in Figure 3. It is structured in 3 different phases. The first one is the *FOM Development* phase. This step, which consists of the automated generation of the FOM, is explained in Section 3.2.1. The *Federate Development* phase deals with the development of a single federate, starting from an event log, which is assumed to be available in XES format and also compliant with the DS Extension. This step is described hereafter, from Section 3.2.2 to Section 3.2.4. The Federate Development phase is carried out through the combined use of the ProM Framework (<https://promtools.org>) and a set of automated transformation steps implemented in Java and Acceleo (<https://www.eclipse.org/acceleo>), the implementation of the MOFM2T OMG standard for the specification of *model-to-text* transformations provided by the Eclipse Framework (<https://www.omg.org/spec/MOFM2T>).

It should be underlined that currently, any ProM plugin is able to understand the additional data provided by the DS Extension to discover a BP collaboration model where various pools represent the different participants. On the other hand, various existing plugins, for example, the Inductive Visual Miner [13] used in this work, can retrieve the additional data and export them as a CSV file. Therefore, the proposed approach introduces automated transformation components that complete the information generated by ProM to produce the required collaboration model, as illustrated in Figure 3.

Finally, it is worth noting that the use of a different PM tool does not affect the validity of the proposed approach. The next sections provide a detailed description of the various activities that constitute the proposed method.

3.2.1 Specification of the Federation Data Model and FOM Development

As aforementioned, it is assumed that a UML data model is available to describe the various entities (e.g. participants and data elements exchanged) involved in the BP collaboration. Therefore, federation requirements are initially identified and then used to define the federation data model and the FOM. Specifically, the following actions are executed.

- The federation requirements are provided to an HLA expert in charge of specifying the UML-based federation data model, which describes:
 - (i) the federation entities and their semantics according to the HLA terminology (e.g., a Federate, an Object Classes, an Interaction Classes, etc.);
 - (ii) the capabilities each participant shall provide (e.g., who publishes/subscribes what);
 - (iii) other HLA-related information required for describing the federation (e.g., synchronization points).

In this respect, federation model specification makes use of the the FOM-HLA Profile [14] that provides appropriate stereotype characterize model elements in terms of HLA-related concepts.

- The first activity of the method shown in Figure 3 is carried out: through the execution of a *model-to-text* transformation, the FOM is generated from the annotated federation model. This step has been introduced in [14] and will not be further discussed in this work.

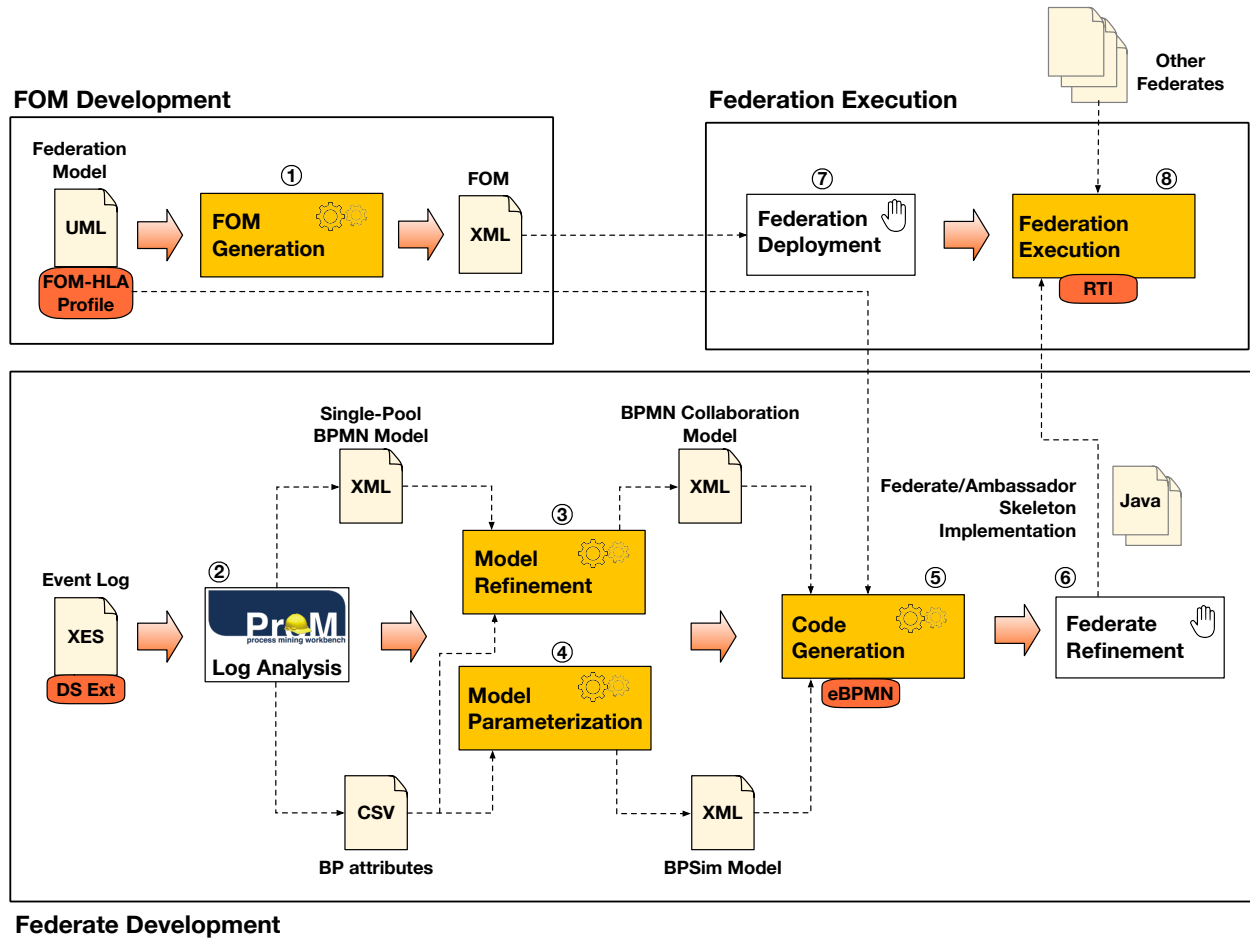


Figure 3: DS- and PPM-based analysis of process collaborations.

3.2.2 Event Log Analysis and Discovering of the Single-Pool BPMN Model

The development of a BP collaboration model that describes the collaborative workflow under study is a two-phase process. The first phase consists of the analysis of the event log that records the activities performed by a single participant (see step 2 of Figure 3). In the proposed method this step is carried out by use of the ProM Tool. The objective of this analysis is to discover:

- a preliminary BPMN model that describes the actual flow of activities carried out by the participant under study;
- all the parameters that quantitatively specify the process flow:
 - *workload characterization*, by discovering the probability distribution function (PDF) that best represents the *inter-arrival time* between two requests;
 - *task duration*, which is represented as a random variable. The log analysis is thus conducted to discover, for each task, the PDF that best fits with the duration data recorded in the event log;
 - the *resource characterization*, in terms of *roles* which group the actual task executors (e.g., human operator or automated resources such as IT systems or local or remote services) along with the relevant *cardinality* (e.g., the number of physical different resources playing a given role);
 - *routing probability*, to be assigned to each gateway in the BP model.

The following two items constitute the output of this step:

- a single-pool BPMN model specifying the process executed by the participant. The model is serialized to an XML file;
- a CSV file, exported by the ProM Inductive Visual Miner plugin, which provides various performance-related parameters about the execution of single tasks and the entire process, along with the additional information provided by the DS Extension.

A preliminary discussion of methods, tools, and technologies that could be adopted to perform this step has been provided in [4]. Such a contribution does not assume the use of any XES extension that addresses distributed systems and thus does not take into consideration the possible use of distributed simulation. Differently, in this work, the system log is assumed to comply with the proposed *DS Extension*. As a consequence, the analysis conducted in the second step is able to identify the activities that send/receive data to/from other participants.

Finally, it is worth noting that the analysis might benefit from pre-processing the event log to assess the quality of the recorded data. Additionally, filtering the event log using a frequency-based approach may be necessary to exclude incomplete traces and/or extremely rare activities to reduce log noise [12, 4]. Such preliminary actions, which may precede the discovery of the BP model, do not affect the principles underlying the proposed method. Interested readers are referred to the relevant literature, as this aspect is not further addressed in this work.

3.2.3 BPMN Model Refinement and Parameterization

The specification of the BPMN Collaboration model and the identification of relevant performance-related parameters that quantitatively characterize the model are dealt with in the Model Refinement and Model Parameterization activity, respectively (see 3 and 4 in Figure 3).

Specifically, the single-pool BPMN model and the CSV file are given as input to the *Model Refinement model-to-text* transformation, which produces as output the BPMN Collaboration model. Such a model includes:

- A BPMN `Pool` element representing the participant performing the log analysis. This process is fully specified, and its activity flow is directly generated from the single-pool model provided as input to the transformation. Furthermore, all BPMN `Task` elements corresponding to events where the `CommType` attribute is not null, are translated to *Send Task* or *Receive Task*, accordingly;
- An empty BPMN `Pool` element for each different entity specified by the additional `RemoteEntity` attribute included in the CSV file;
- an appropriate set of *BPMN Message* element, which is sent to or received by *Send Task* and *Receive Task* element, respectively. Specifically, messages originate from (or are directed to) the empty pool according to the value of the additional `RemoteEntity` attribute of the corresponding log event. Finally, *BPMN Message* element refers to the value assumed by the additional `Data` attribute associated with the related event.

The approach described above improves the limitations of the previously proposed method [9] by making no longer necessary the manual refinement previously required to specify the exchanged message for each Send / Receive task in the model.

The parameters of the BPMN Collaboration are specified by use of the Business Process Simulation Interchange Standard (BPSim), the WfMC standard for the specification and interchange of process analysis data (<https://www.bpsim.org/>). In this respect, in step 4, the CSV file is given as input to the *ModelParametrization* model-to-text transformation, which generates as output a BPSim file describing the parameters needed for configuring the simulation model. It is worth noting that the use of separated software items to provide the BPMN model and the corresponding parameterization brings the following advantages

- *Separation of concerns*: the structural and topological description of the simulation model is separated from its parameterization. Both make use of standards to ensure compliance with third-party tools that might be introduced to complement and extend the proposed approach's capabilities.
- *Decoupling of process model and analysis strategy*: this work proposes the use of simulation to analyze the BP collaboration and specifically adopts Java, HLA, and the eBPMN framework. Different simulation environments and technologies, as well as different analysis approaches, might be adopted to process the BPMN/BPSim models to address different domain-specific needs while preserving the overall validity of the proposed strategy.

The BPMN Collaboration, together with the performance-related characterization provided by the BPSim model, constitutes the *simulation model* of the collaborative scenario under study.

3.2.4 Federation Implementation, Deployment and Execution

At step 5, the federate implementation is developed by executing the *Code Generation* component. The proposed method, based on the strategy introduced in [9], uses a specific component for the automated generation of the implementation. This component takes as input the simulation model and generates as output the corresponding skeleton of the eBPMN/HLA federate implementation, which is refined and completed at step 6 (*Federate Refinement*).

Finally, in steps 7 and 8, the federate is deployed along with the federates developed by other participants, and the federation is executed to simulate the required scenario.

The next Section provides an example application that illustrates how the proposed method can be used in practice to support the distributed simulation of a process collaboration scenario.

4 EXAMPLE APPLICATION

The example application deals with a purchasing process that involves two actors: a retailer and a supplier. The process is structured as follows: the retailer collects purchase requests from end customers and forwards them to the reference supplier for the provisioning of the requested goods. The supplier also handles the shipping of the requested goods to the customer. For certain product types, the supplier is also responsible for retrieving and assembling the necessary components before shipping the purchased items.

One of the objectives of the proposed approach is also to demonstrate how, through the use of an appropriate toolchain, the various participants involved in the collaborative process can benefit from distributed simulation technologies and a low-code development paradigm. Each participant has limited knowledge of the whole process and has no visibility of the internal processes implemented by other partners. The development of a simulation system in a distributed environment is generally a challenging task that typically requires significant expertise in DS standards and technologies. To mitigate this complexity, an automated tool that minimizes manual activities would be highly beneficial. The next sections outline how the analysis has been carried out.

4.1 Specification of the Federation Object Model and FOM Generation

In the first step, the federation data model is specified by the use of a UML class diagram annotated with the FOM-HLA profile. The annotated model is shown in Figure 4. The related FOM, obtained as the output of the *FOM Generation* transformation, is omitted for the sake of brevity.

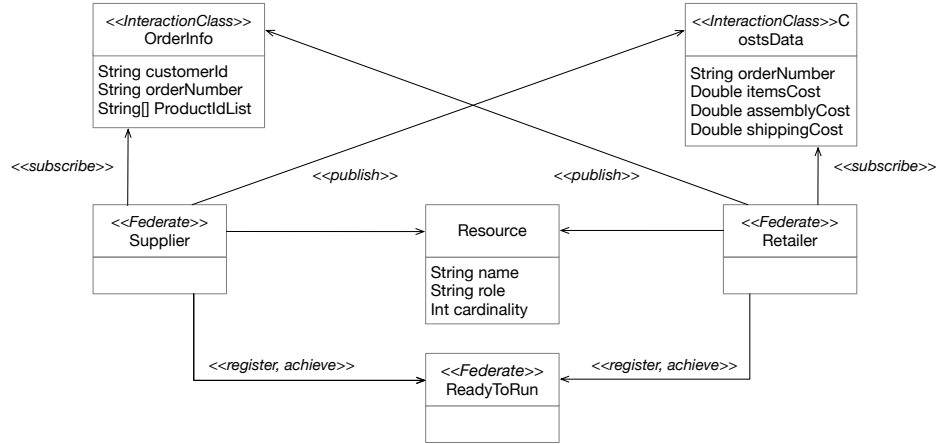


Figure 4: Federation Data Model.

4.2 PM-based Analysis of Participants' Event Log

Both the *Supplier* and *Retailer* participants analyze their event logs to discover the single-pool model, which describes the flow of activities performed on each partner's side. Figure 5 shows a fragment of the supplier log. For each event, the log contains the standard attributes that identify a *trace* and the *activity name*. In addition, there are attributes provided by three standard extensions [15]: *Start Timestamp* and *Completion Timestamp* from the *life-cycle* and the *time* extensions, *Resources* and *Role* from the *organizational* extension. Finally, the log also includes the attributes specified by the DS Extension: *Local Entity* and *Remote Entity*, *Communication Type* and *Data*. It is important to underline that, to date, no PAIS are capable of capturing the data required by the DS Extension. Therefore, for the case study, synthetic logs were used, generated through an additional feature of the eBPMN framework, specifically developed in the context of this research.

The *IDV* plugin, used to analyze the above-mentioned event log, allows the discovery of a process model specified as a Petri Net, as depicted in Figure 6. The latter has been transformed to a single-pool BPMN model throughout the *Convert Petri Net to BPMN* plugin [16] and finally exported. With regard to the discovery of simulation parameters, the IDV results have also been exported in CSV format.

Caseld	StartTimestamp	CompletionTimestamp	Activity	Resource	Role	LocalEntity	RemoteEntity	CommunicationType	Data
1	2025/01/09 08:32:29.085	2025/01/09 08:42:09.344	Receive Order Details	Li Wei	Supplier Officer	Supplier	Retailer	receive	OrderInfo
1	2025/01/09 08:42:09.344	2025/01/09 09:02:31.225	Retrieve Components	Mario Rossi	Production Worker	Supplier			
2	2025/01/09 08:54:58.745	2025/01/09 09:04:52.175	Receive Order Details	Li Wei	Supplier Officer	Supplier	Retailer	receive	OrderInfo
1	2025/01/09 09:02:31.225	2025/01/09 09:32:47.768	Assembly Items	Mario Rossi	Production Worker	Supplier			
2	2025/01/09 09:32:47.768	2025/01/09 09:56:21.093	Retrieve Components	Mario Rossi	Production Worker	Supplier			
1	2025/01/09 09:32:47.768	2025/01/09 09:35:00.329	Send Supply Costs Details	Li Wei	Supplier Officer	Supplier	Retailer	send	CostsData
1	2025/01/09 09:35:00.329	2025/01/09 09:35:55.586	Ship Items	Li Wei	Supplier Officer	Supplier			

Figure 5: Supplier Event Log (Excerpt)

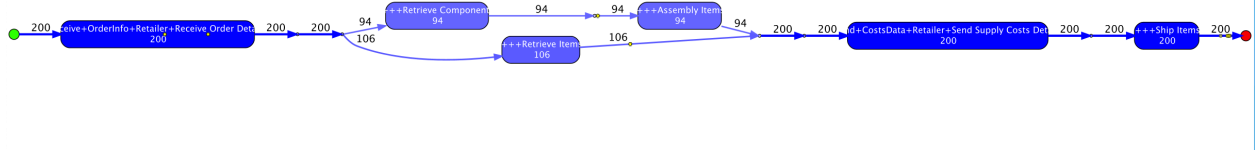


Figure 6: PetriNet discovered by the *Inductive Visual Miner (IVM)* plugin.

4.3 BPMN Model Refinement and Parameterization

The *Model Refinement* transformation takes the single-pool model and the CSV file as input to generate the collaboration model, as described in Section 4.3. The resulting BPMN Collaboration obtained by analyzing the supplier log is provided in Figure 7.

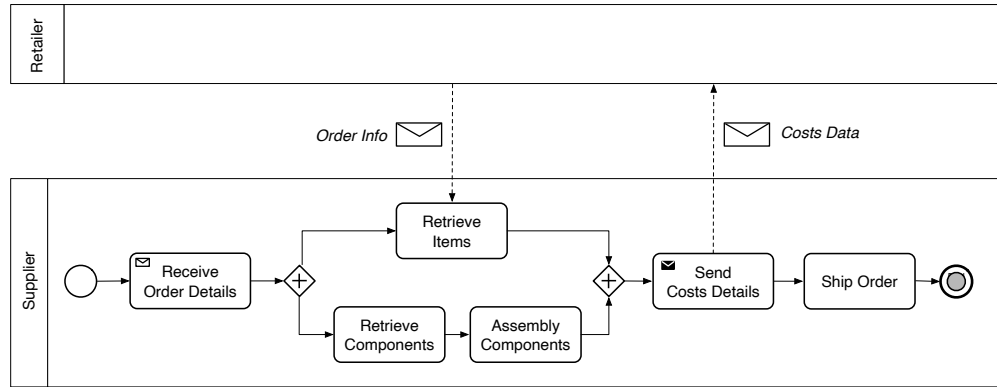


Figure 7: BP Collaboration from the perspective of the *Supplier* participant.

The role of the Model Parameterization component is to generate the BPSim model, which provides the performance-related parameters that complete the BP Collaboration model. Notably, some parameters are directly generated by the IVM plugin, while others require to be computed by processing and aggregating data contained in the CSV file. Specifically:

- *Process Workload*: in the adopted simulation model, the inter-arrival time of purchasing requests is modeled through a random process. As observed in [12], a negative exponential distribution can realistically be assumed for the interarrival process. The *Model Parametrization* component estimates the intensity parameter λ .
- *Task Service Time*: according to [12] a normal distribution is assumed to model the service time, where μ and σ are estimated using the mean value and variance of the task service time. These parameters are discovered by the IDV ProM plugin and included in the CSV file.
- *Resource Cardinality and Routing Probabilities*: the number and type (role) of resources involved in each activity execution is discovered by the IDV plugin and exported to the CSV file. Similarly, IDV determines the number of occurrences of task execution in each trace. Based on this information, the *Model Parametrization* component determines the number and the cardinality of each required resource and the routing probabilities of each exclusive diverging gateway.

4.4 Federation Implementation, Deployment and Execution

The federate implementation has been based on the eBPMN framework, which enables the seamless execution of a BPMN model both as a local simulation and as a distributed HLA-based simulation.

The simulation results are summarized in Table 1. For each activity, the average value of the service time and the waiting time obtained from the simulation execution are compared with relevant data discovered throughout the log analysis.

A similar comparison is shown Table 2 with respect to the time elapsed from the supplier process (average, minimum, and maximum values).

Such comparisons show how the obtained simulation model is able to appropriately replay the behavior of the previous process execution, as recorded in the event logs. Therefore, it constitutes a valuable tool for conducting a simulation-based analysis of the process collaborations to predict future behavior and investigate the impact of different operational configurations.

Table 1: Comparison between the average values of activities' service time and waiting time (simulation output vs log analysis). Times are expressed in hours (h), minutes (m) and seconds (s).

Task Name	Simulation Results		Log Analysis	
	Service Time	Waiting Time	Service Time	Waiting Time
Receive Order Details	10m12s	0m00s	10m02s	0m00s
Retrieve Items	18m12s	20m10s	19m52s	19m28s
Retrieve Components	24m13s	1h44m12s	23m45s	1h34m40
Assembly Items	35m09s	2h36m01s	35m07s	2h32m37s
Send Costs Details	1m42s	1h07m07s	1m59s	1h09m07s
Ship Items	1m10s	1h02m08s	1m00s	1h01m52s

Table 2: Comparison between the mean, minimum and maximum value of the elapsed time for the Supplier process (simulation output vs log analysis). Times are expressed in hours (h), minutes (m) and seconds (s).

		Simulation Results	Log Analysis
Elapsed Time	AVG	5h32m59s	5h20m27s
	MIN	31m12s	30m53s
	MAX	14h49m12s	17h30m06s

5 CONCLUSIONS

This paper has shown how the integration of Process Mining and Distributed Simulation effectively addresses the challenges of analyzing collaborative business processes. By leveraging a low-code development paradigm and the XES extension mechanism, the proposed method provides a practical approach to managing complexity and preserving data confidentiality among participants.

Future developments include implementing the prototype tool as a unified ProM plugin to enhance usability. A more challenging direction involves introducing Predictive Process Mining (PPM) in the context of Digital Twins. This would require investigating how the BP collaboration model can incorporate real-time measurements from the process it represents and, conversely, how it can become adaptive by evolving based on simulation outcomes.

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